

PEDRO FISCHETTI

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Education

University of South Carolina

Columbia, SC

Bachelor of Science in Computer Engineering, GPA: 3.20

Dec 2025

- Relevant Coursework: Data Structures & Algorithms, Embedded Systems, Signal Processing, Digital Logic Design

USCG Federal Maritime Boarding Officer School

North Charleston, SC

Secret Security Clearance (Active until 2027)

Dec 2020

Related & Professional Experience

United States Coast Guard

Various Locations

Machinery Technician 2nd Class (Active Duty)

Feb 2016 – Dec 2023

- Sustained mission-critical engineering and electronic systems aboard cutters, ensuring operational reliability
- Managed the safe handling and transfer of hazardous materials during 2018 dry-dock availability (8,000 gallons fuel oil, 330 gallons lube oil, 660 gallons coolant)
- Diagnosed and resolved critical equipment failures (heat exchangers, pumps, regulators), ensuring mission continuity
- Collaborated on navigation aid maintenance across Chesapeake and Delaware Bay, supporting vital marine transports
- Demonstrated technical proficiency beyond rank, overcoming shipboard discrepancies and ensuring seamless operations

Best Buy

Coral Springs, FL

Connections Manager

Aug 2013 – Dec 2015

- Led 70+ member team through major industry transition, consistently exceeding \$100K daily sales targets
- Developed comprehensive training programs that improved team performance and customer satisfaction

Project Experience

ArduinoGotchi - Tamagotchi Emulator | C++, Arduino UNO, SSD1306 OLED

2024

- Developed full P1 emulator under strict memory constraints (32KB Flash, 2KB RAM)
- Implemented custom Hardware Abstraction Layer and optimized LCD rendering with bit manipulation
- Integrated Java-based ROM conversion utility for cross-platform compatibility

IoT Multi-Display Task Manager | ESP32, WiFi, HTTPS, JSON APIs

2024

- Engineered an IoT system managing three SSD1306 displays via I2C multiplexers
- Implemented secure HTTPS API integration for real-time task synchronization and dynamic UI

Home Server & IoT Hub | Raspberry Pi, Docker

2024

- Built local server with Home Assistant, live notes, vegan cookbook, and educational game containers

3D Printed Desktop Buddy | Fusion 360, Bambu Studio

2024

- Designed and printed custom shells integrating soldered multi-display modules for dynamic interaction

Healthcare Automation Script | Bash, CSV Parsing

2024

- Automated staff competency tracking and compliance reporting, reducing manual workflows

Technical Skills

Languages: C++, Python, JavaScript/Node.js, MATLAB, Java, Bash

Embedded Systems: Arduino (UNO, ESP32, Nano, Mega), I2C/SPI/UART, memory optimization

Tools: Arduino IDE/CLI, Git, NVM, Zsh, Fusion 360, Bambu Studio, LTSpice, Docker

Frameworks: Adafruit (SSD1306, GFX, BusIO), U8g2, WiFiManager, ArduinoJson, Google Test

Specialties: Digital logic design, signal processing, IoT development, 3D design, real-time systems

Languages: Portuguese (Native), Spanish (Conversational), English (Fluent)

Leadership Experience

USCG Law Enforcement Operations

Certified Instructor & Firearms Coach | Maritime Enforcement Leadership

2016 – 2023

- Directed complex maritime enforcement operations, recognized with multiple awards for operational excellence
- Enhanced team operational readiness through advanced training program development